

Classic machine learning on top of multiple position weight matrices improves genomic prediction of transcription factor binding sites

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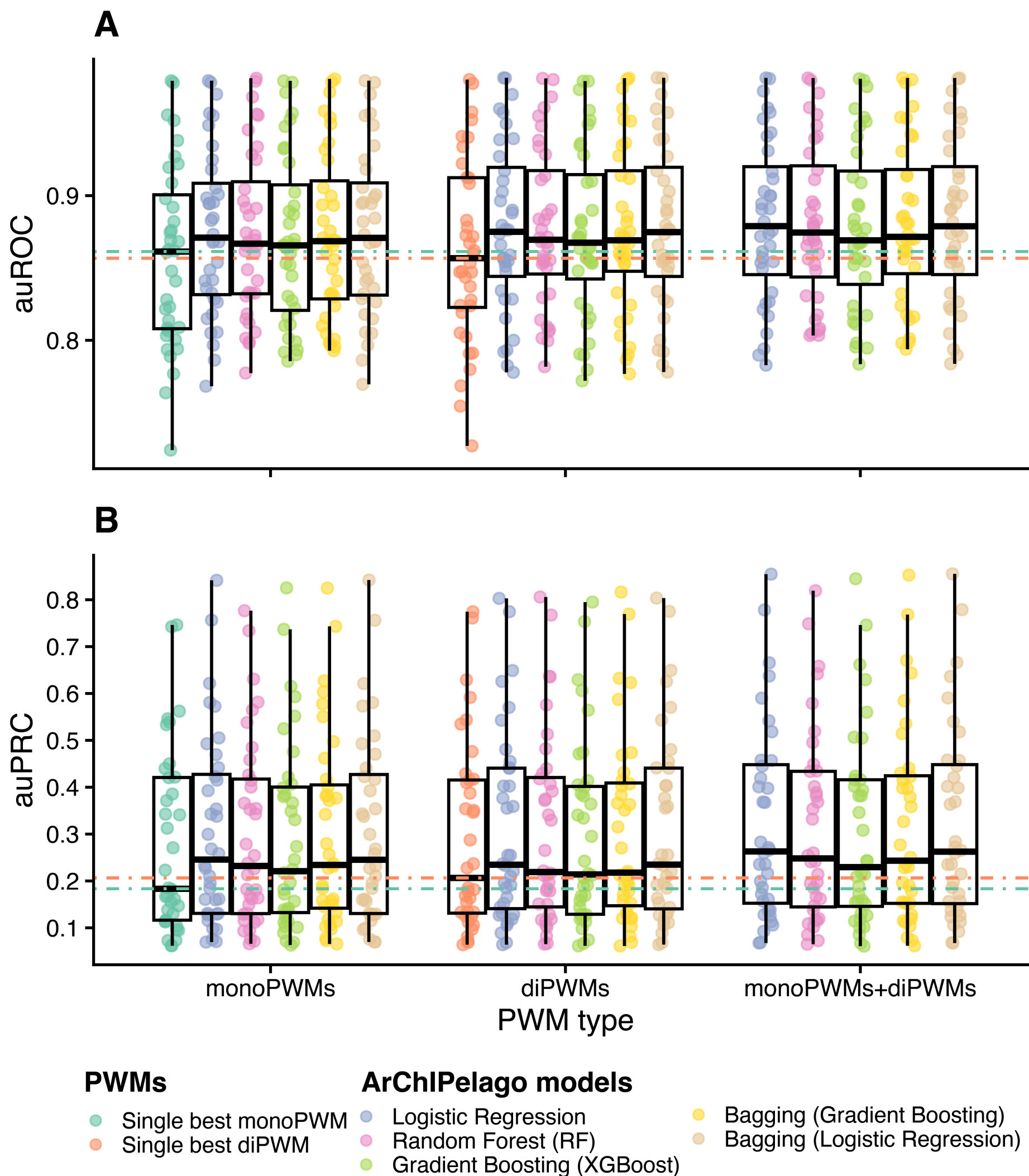


Figure S1. Integrative ArChIPelago models outperform individual PWMs on the mouse test set. The models were trained on human data and tested on mouse data. Random Forest (RF), Gradient Boosting (XGBoost), Logistic Regression, and ensemble Bagging consistently achieve superior performance compared to both best monoPWMs (green horizontal line: median across TFs) and best diPWMs (orange horizontal line: median across TFs). Each dot represents a model performance for a single TF (A is auROC, B is auPRC).

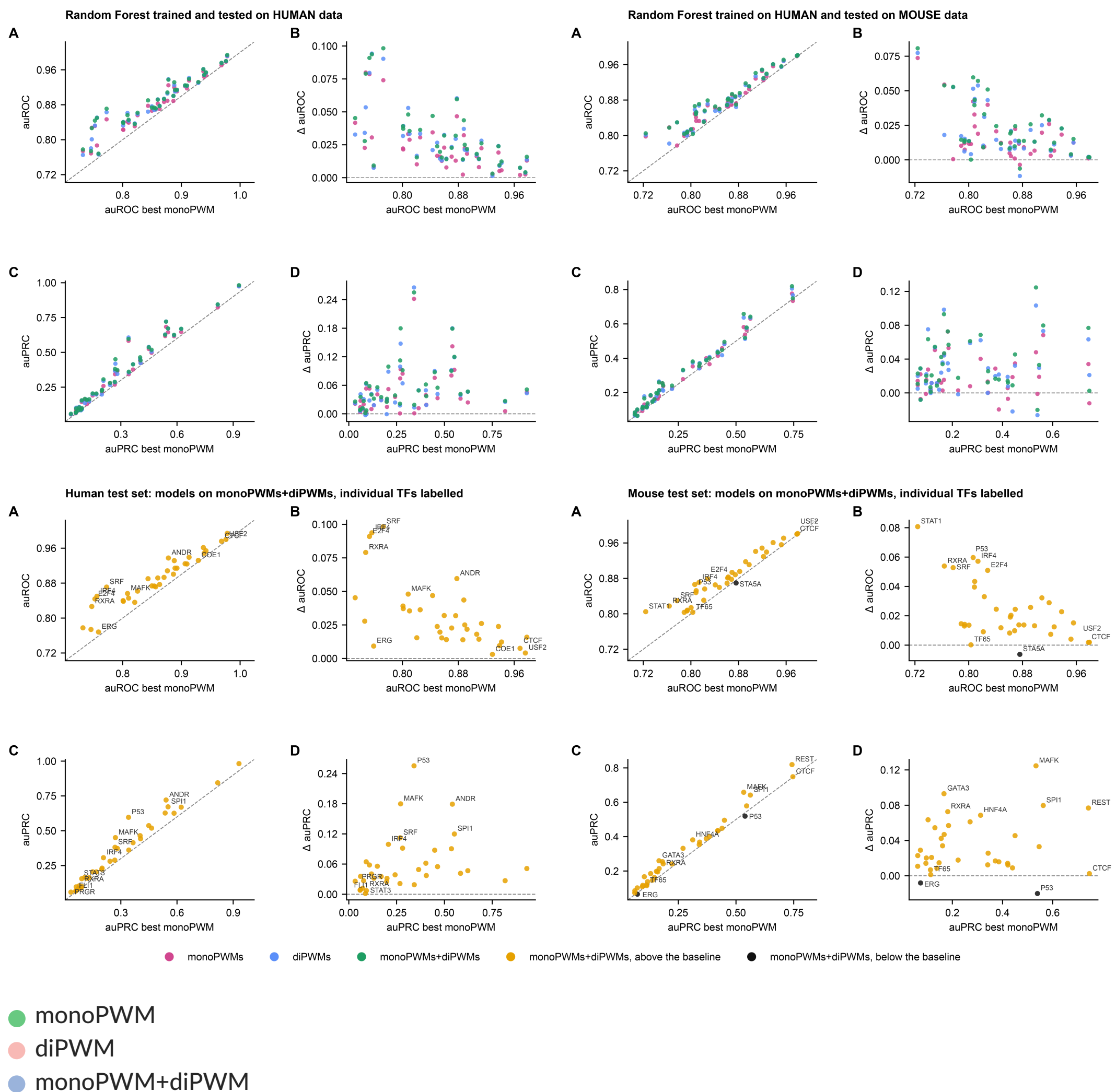


Figure S2. Performance of the Random Forest trained and tested on human data (left half) and trained on human data but tested on mouse data (right half). A, B show auROC values and their differences (Δ auROC), C, D show auPRC values and their differences (Δ auPRC). Results are displayed for models trained with monoPWMs, diPWMs and combined monoPWMs and diPWMs (upper block, coloured by PWM set). Individual transcription factors are labelled in the bottom block (models on monoPWMs and diPWMs), illustrating cases with improved or reduced cross-species generalisation. The reference (x axis) is the best single monoPWM of the TF, selected on the human training set, for all three model types; TFs above this reference are shown in amber and TFs below it in black.

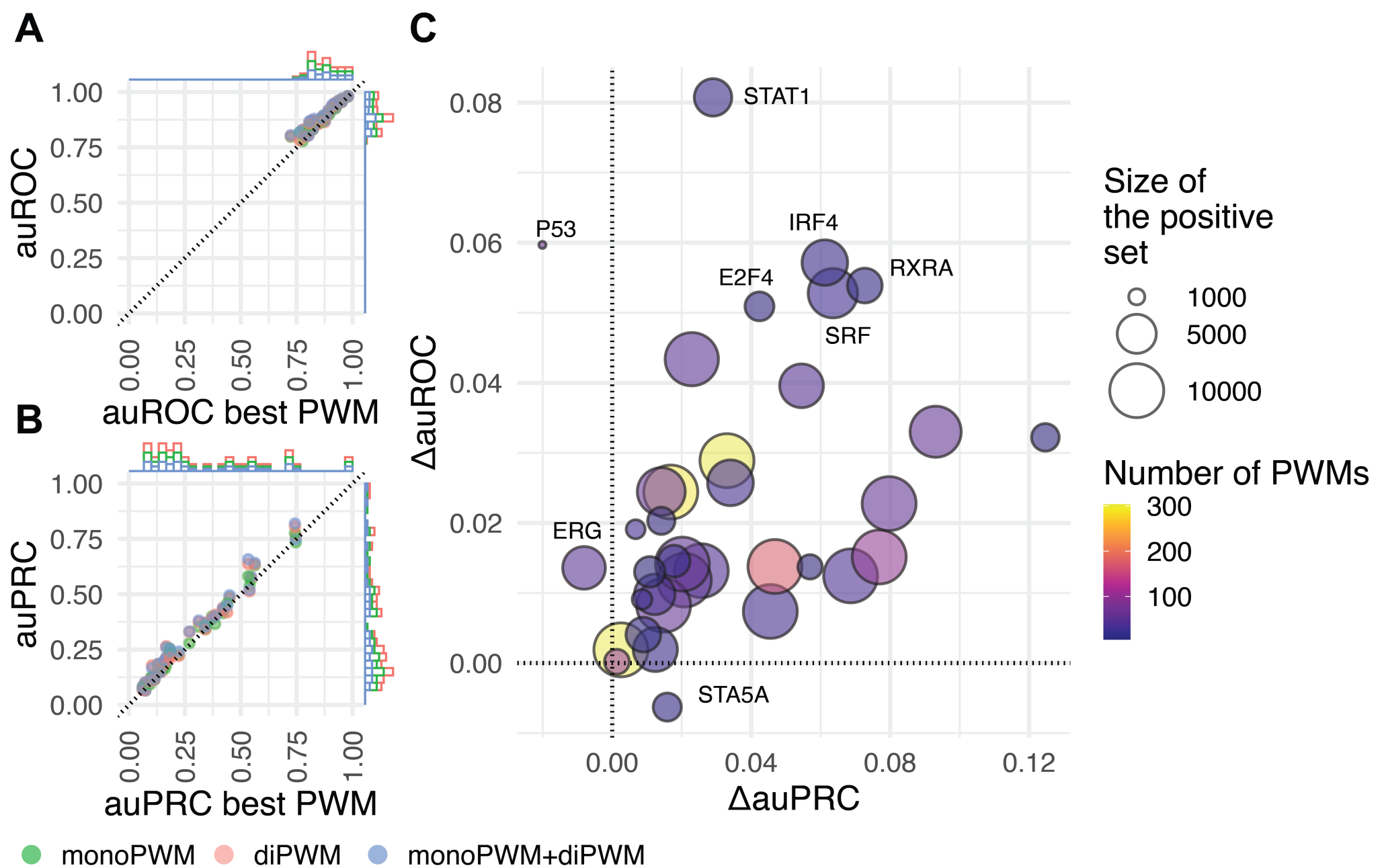


Figure S3. ArChIPelago outperforms the best PWMs by accounting for different motif subtypes: mouse test set. The models were trained on human data and tested on mouse data. A. An overall auROC improvement for models built with several PWMs (monoPWMs, diPWMs, or both) in comparison to the best single monoPWM selected on the human training set (Methods). Three coloured dots per TF are shown; the histograms along the axes show the distributions. B. An overall improvement of auPRC: models built with several PWMs in comparison to the best single monoPWM. C. The difference in auROC and auPRC between the ArChIPelago model (trained on both monoPWMs and diPWMs) and the individual best monoPWM. Dot colour gives the total number of monoPWMs and diPWMs of the TF and dot size the number of its training positives; the TFs with the largest and the smallest gains are labelled.

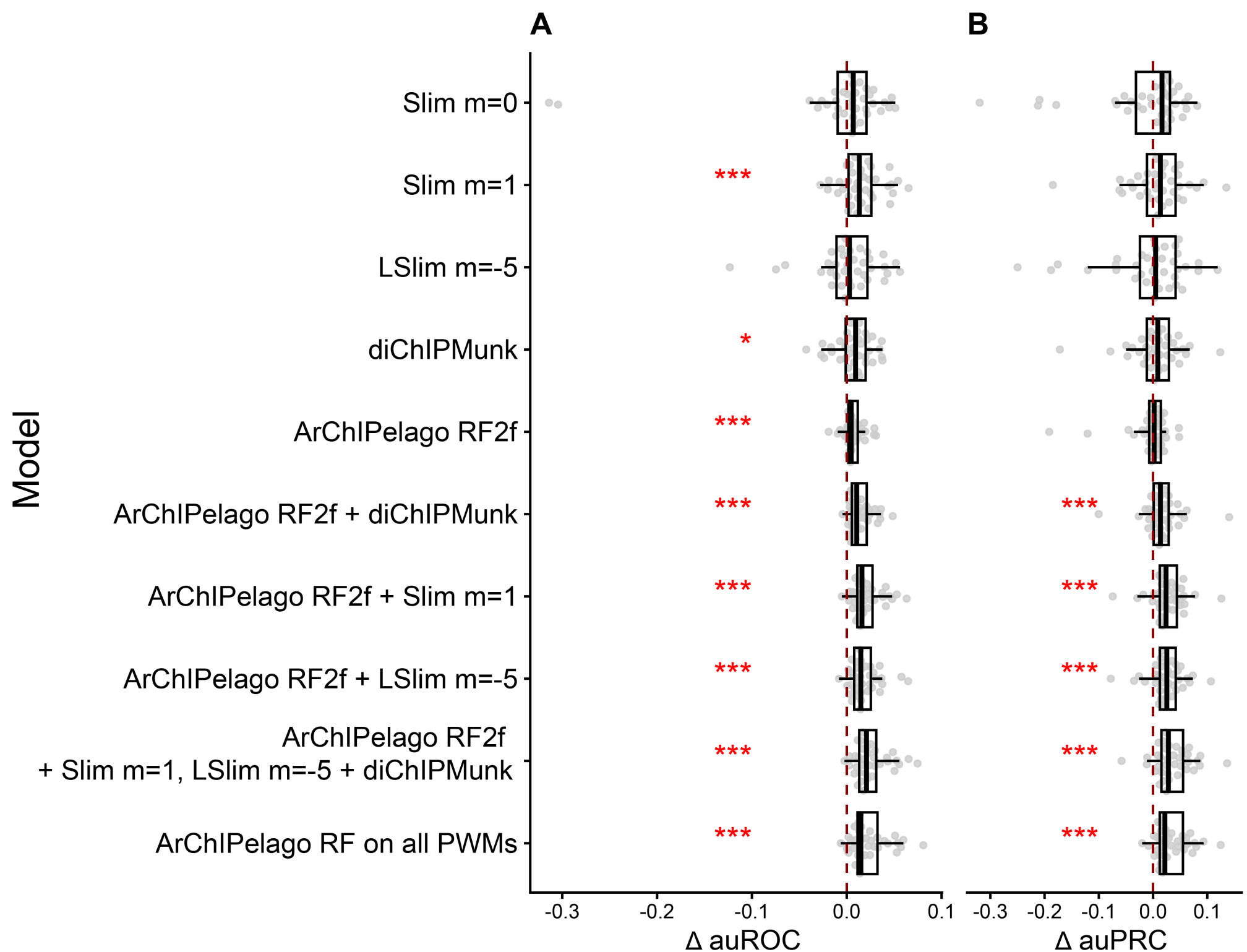


Figure S4. ArChIPelago outperforms diPWMs and Slim models of different degrees on the mouse test set. The models were trained on human data and tested on mouse data (mouse chromosomes 1, 8 and 19). The comparison of ΔauROC (A) and ΔauPRC (B) between the best monoPWMs and individual models, namely Slim m=0, Slim m=1, LSlim m=-5, diChIPMunk diPWM (built from all the positive genomic sequences), ArChIPelago RF on the best monoPWM and the best diPWM (RF2f, two features only), RF2f with the above-mentioned models included as additional features, and ArChIPelago RF on all monoPWMs and diPWMs of the TF (RF on all PWMs). Each dot is one of the 36 TFs; the reference (dashed line at zero) is the best single monoPWM of the TF, selected on the human training set and evaluated on the mouse test set (Sup. Table 3). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, one-sided Wilcoxon signed-rank test of the per-TF differences against zero.

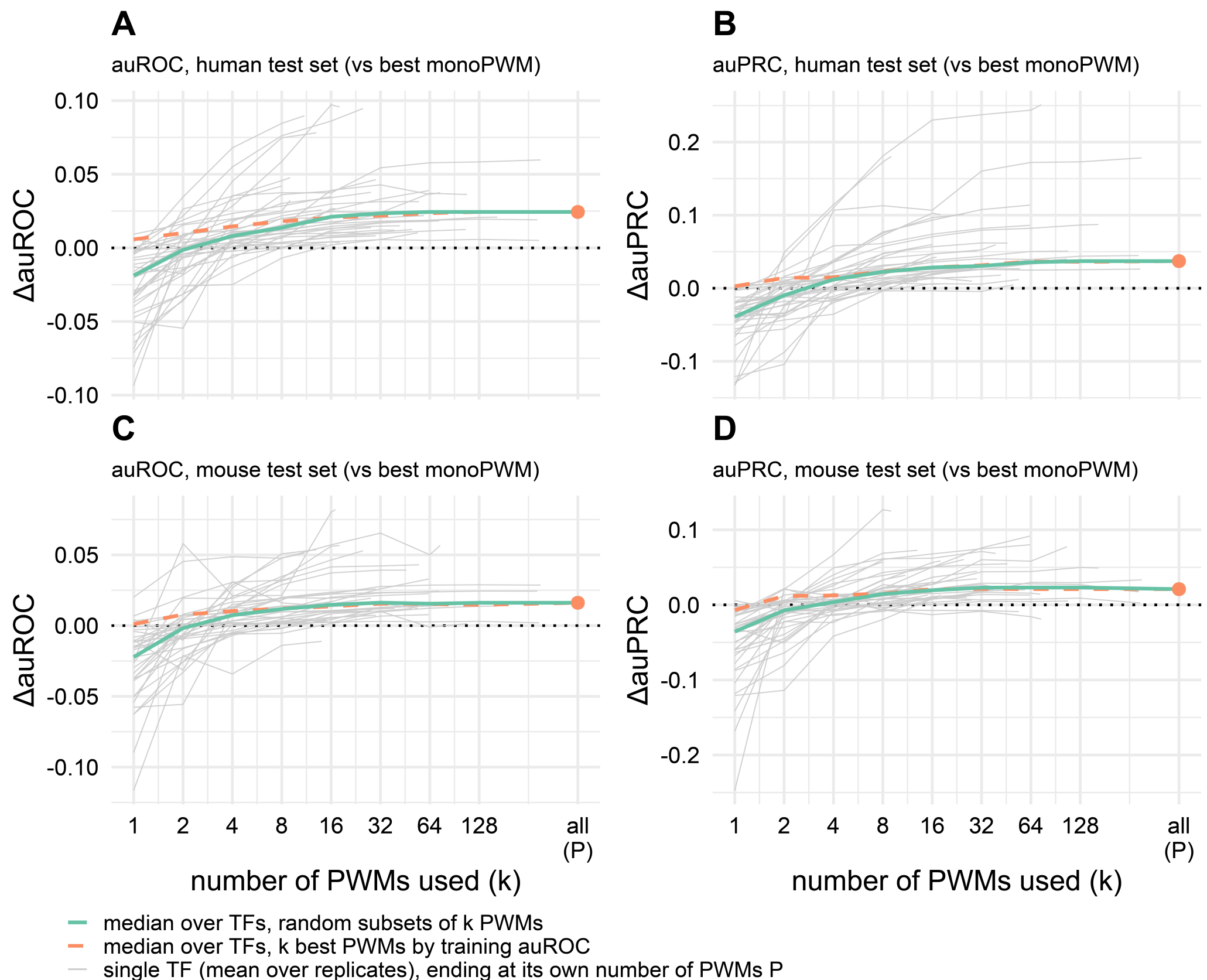


Figure S5. Performance of ArChIPelago as a function of the number of PWMs used. For each of the 36 TFs the Random Forest was re-trained on random subsets of $k = 1, 2, 4, \dots, 128$ of its monoPWMs and diPWMs (five subsets per k) and, separately, on the k PWMs with the highest auROC on the training set; the full model (all P PWMs of the TF) is shown at the right of each panel. Performance is expressed as the difference from the best single monoPWM selected on the human training set. A, auROC on the human test set; B, auPRC on the human test set; C, auROC on the mouse test set (mouse chromosomes 1, 8 and 19); D, auPRC on the mouse test set. Grey lines: individual TFs (mean over replicates), each ending at its own number of PWMs P . Green line: median over the 36 TFs for the random subsets; orange dashed line: median for the top- k subsets. A TF with $P < k$ enters at $k = P$, so every median is computed over all 36 TFs and gives the gain obtained with a budget of k PWMs.

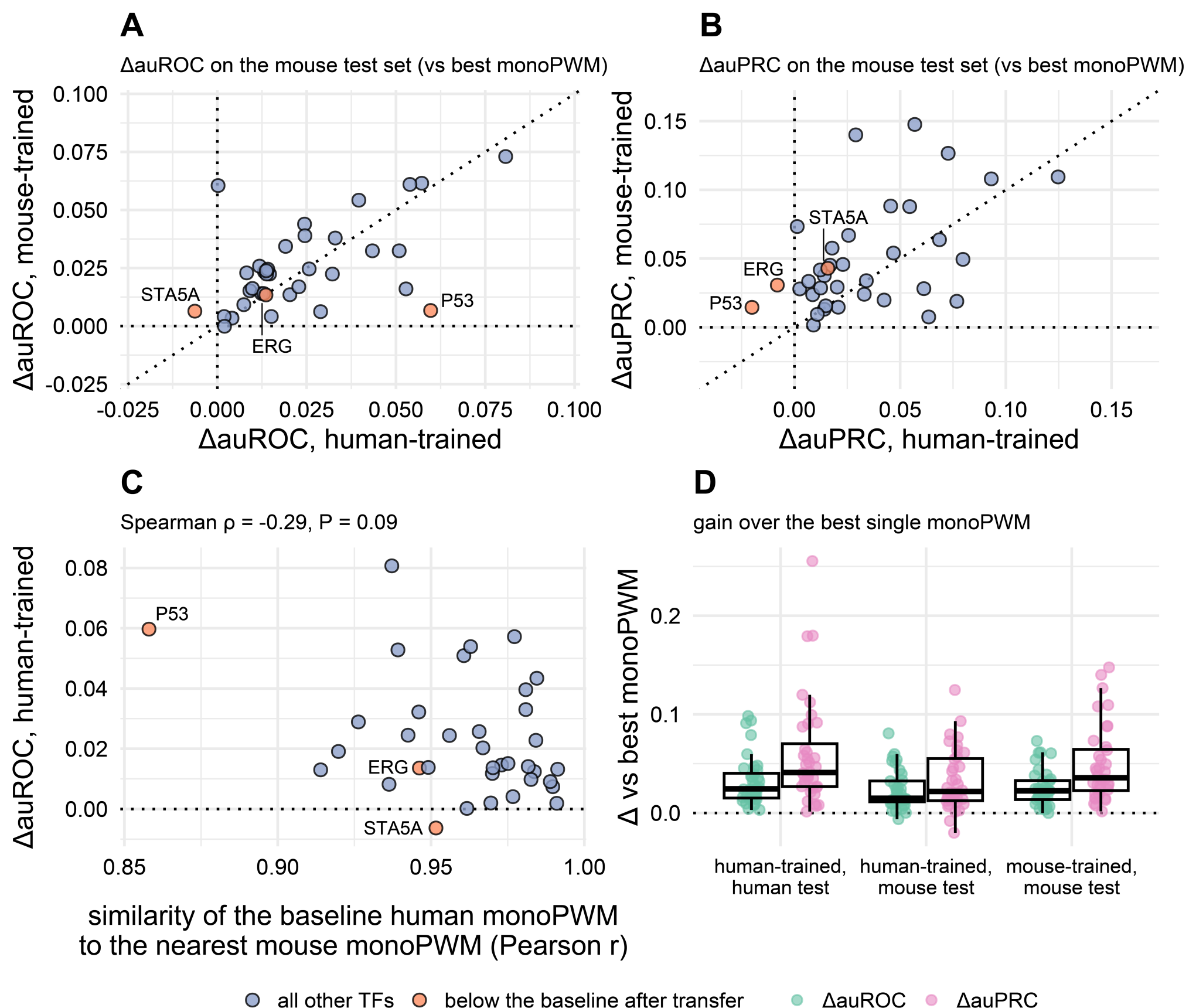


Figure S6. Cross-species transfer and the mouse-trained control. A, B: per-TF gain over the best single monoPWM of the human-trained model evaluated on the mouse test set (x axis; baseline: the best single monoPWM selected on the human training set) against the gain of the mouse-trained model on the same mouse test chromosomes (y axis; baseline: the best single mouse monoPWM selected on the mouse training set), for auROC (A) and auPRC (B). Orange: the three TFs that fall below the baseline after transfer on at least one metric; all three are above the baseline when the model is trained on mouse data. C: the cross-species gain in auROC against the similarity between the baseline human monoPWM (the best by training auROC) and the most similar mouse monoPWM of the same TF (maximum over all mouse monoPWMs of the TF and over alignments of the mean per-column Pearson correlation of the log-odds matrices; Spearman correlation given above the panel). D: distribution over the 36 TFs of the gain over the best single monoPWM in the three settings (human-trained on the human test set, human-trained on the mouse test set, mouse-trained on the mouse test set), in green for auROC and in pink for auPRC; boxes show the median and the interquartile range, whiskers extend to the most extreme value within 1.5 interquartile ranges, dots are individual TFs.

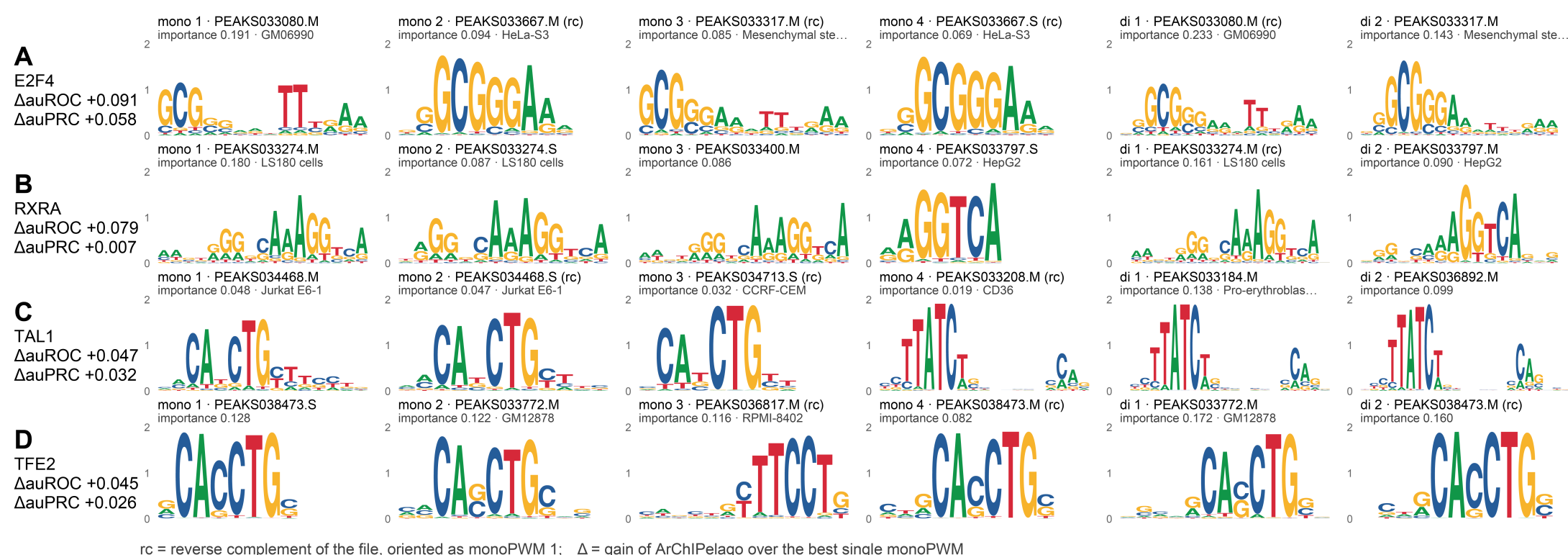


Figure S7. Motif subtypes captured by ArChIPelago. For E2F4 (A), RXRA (B), TAL1 (C) and TFE2 (D), four of the eight TFs with the largest auROC gain of the Random Forest over the best single monoPWM on the human test set, the four monoPWMs (mono 1 to 4) and the two diPWMs (di 1, 2) with the highest Random Forest feature importance in the model trained on monoPWMs and diPWMs are shown as sequence logos (information content in bits). The feature importance, the ChIP-Seq peak set and, where available, the cell type are given above each logo; rc marks a PWM drawn as the reverse complement of the file so that all logos of a TF are oriented as its first monoPWM; diPWMs are drawn as their mononucleotide marginal. Δ auROC and Δ auPRC on the left: gain of ArChIPelago over the best single monoPWM on the human test set. For each TF the top-ranked PWMs represent distinct motif subtypes: the canonical E2F site and an E2F site followed by a TTTGAA element at a fixed spacing (A); the DR1 direct repeat and the single AGGTCA half-site (B); the E-box of T-cell lines and the GATA-E-box composite element of erythroid cells (C); the palindromic CAGCTG E-box and a CTTCCTG-containing element (D).

Supplementary tables

Sup. Table 1. ChIP-Seq experiments. The 414 human and 290 mouse ChIP-Seq experiments used in the study (sheets 'human' and 'mouse'). For every TF, the GTRD peak set, experiment and control experiment identifiers, the TFClass and UniProt identifiers and the number of GEM peaks.

Sup. Table 2. PWMs and datasets per TF. For each of the 36 TFs, the numbers of monoPWMs and diPWMs, ChIP-Seq experiments, pooled peaks, training and test positives of both species (human test set: chromosomes 1, 8 and 21; mouse test set: chromosomes 1, 8 and 19) and the mean GC content of the positive and negative sequences of every set (sheet 'per_TF'); sheet 'summary' gives the totals and the medians over TFs.

Sup. Table 3. ArChIPelago performance. auROC and auPRC of every model (five algorithms, three PWM sets, 36 TFs) on the human training set and on the human and mouse test sets, together with the best single monoPWM and diPWM of each TF selected on the human training set (sheet 'performance'). Sheet 'summary' compares the Random Forest trained on monoPWMs and diPWMs with the best single monoPWM on both test sets (medians, per-TF gains, Wilcoxon signed-rank tests).

Sup. Table 4. Operational metrics. For every TF and test set, the ranked predictions of the ArChIPelago Random Forest (monoPWMs and diPWMs) and of the best single monoPWM (selected on the human training set by auPRC): false positives per 10,000 candidate regions and precision at 50 % and 80 % recall, the reduction in false positives and the paired precision gain of ArChIPelago, and the number of true binding sites among the top 1,000 and the top 1 % of ranked regions (sheet 'per_TF'). Sheet 'summary' gives the medians and interquartile ranges over the 36 TFs.

Sup. Table 5. Cross-species transfer and the mouse-trained control. For every TF, the gain over the best single monoPWM of the human-trained model on the human and on the mouse test set (baseline selected on the human training set) and of the mouse-trained model on the mouse test set (mean of two seeds; baseline: the best mouse monoPWM selected on the mouse training set), the TF family (TFClass), the numbers of PWMs, experiments, cell types and positives, and the similarity of the baseline human monoPWM to the most similar mouse monoPWM (sheet 'per_TF'). Sheet 'summary' gives the medians and Wilcoxon signed-rank tests of the comparisons; sheet 'correlations' the Spearman correlations of the cross-species gain with TF properties, including the one shown in Fig. S6C.

Sup. Table 6. Runtime and memory. Hardware, software and benchmark inputs (sheet 'setup'); wall-clock time, CPU time and peak memory of PWM scanning with SPRY-SARUS ('PWM_scanning'), of Random Forest training on the full human training matrices ('RF_training') and of inference for 10^6 sequences ('RF_inference'), measured for the monoPWM sets of STAT3, HNF4A and CTCF (5, 26 and 198 PWMs), single-threaded and with four threads.